



kNN-Classification – Exercise 3

Wheat Seeds Dataset

In this exercise, you need to first setup a Python environment. Even if you do not solve exercise b), you should spend the time to set up this environment, as we will be using it in later exercises as well.

- a) Setup a Python development environment incl. numpy, pandas, scikit-learn, matplotlib.
Recommended: use a docker-based (see <https://docs.docker.com/install/>) environment.

1) Build an image/docker container based on the official `jupyter/tensorflow-notebook` hosted at `quay.io`. Unfortunately, it does not include all libraries we will be using (e.g., XGBoost), thus you will find a `Dockerfile` in the LC, which uses it as a base and installs the missing libraries. Run

```
docker build -t ml .
```

in the directory you saved the `Dockerfile` in to build the `ml` image (do not forget the trailing `.` in the command!)

2) Run the docker container

- As you cannot store documents inside a docker container (containers/images get destroyed all the time), you will have to “map” a persistent directory to store your files in into the container. We will be using the `<class directory>` as persistent directory and map it into the container as `“/home/jovyan/work”` (this path is predefined by the container) with the `-v` option.
- We also need to specify a port for the web-server to listen to, we will be using 8888 (`-p`),
- it makes sense to clean up the filesystem after running the container (`--rm`),
- and to give it a clear name (`--name`).

Examples:

Open a terminal, change into your `<class directory>` and run the container:

- on Linux or MacOS, execute

```
$ docker run -p 8888:8888 -v "$PWD":/home/jovyan/work --rm --name ml ml
```

- on Windows storing your files in `C:\temp\`, execute

```
$ docker run -p 8888:8888 -v C:\temp\:/home/jovyan/work --rm --name ml ml
```

If this works properly, it will show you how to access the Notebook-Server through a web browser, this should look like this (your token will be different every time you start the Notebook-Server):

```
To access the notebook, open this file in a browser:
file:///home/jovyan/.local/share/jupyter/runtime/jpserver-7-open.html
Or copy and paste one of these URLs:
http://ee4d6332eca1:8888/?token=1ae2a6f5cae0fbad59dd8eda6c1c8346bdcd4ddb5055c89
or http://127.0.0.1:8888/?token=1ae2a6f5cae0fbad59dd8eda6c1c8346bdcd4ddb5055c89
```

Copy-and-Paste the last URL (the one starting with `http://127.0.0.1:8888/...`) into a web browser on your computer. You should now see the JupyterLab UI.

In case you do not want to use a docker based environment, you can also use Google Colabs or a local install.

- b) Implement a *k*NN classifier in Python/scikit-learn, using the “Wheat Seeds” Dataset from the UCI ML Repository, available at <https://archive.ics.uci.edu/ml/datasets/seeds>